

The Fundamentals of Programming

Functions and Arithmetic Operators



Objectives

What you will learn:

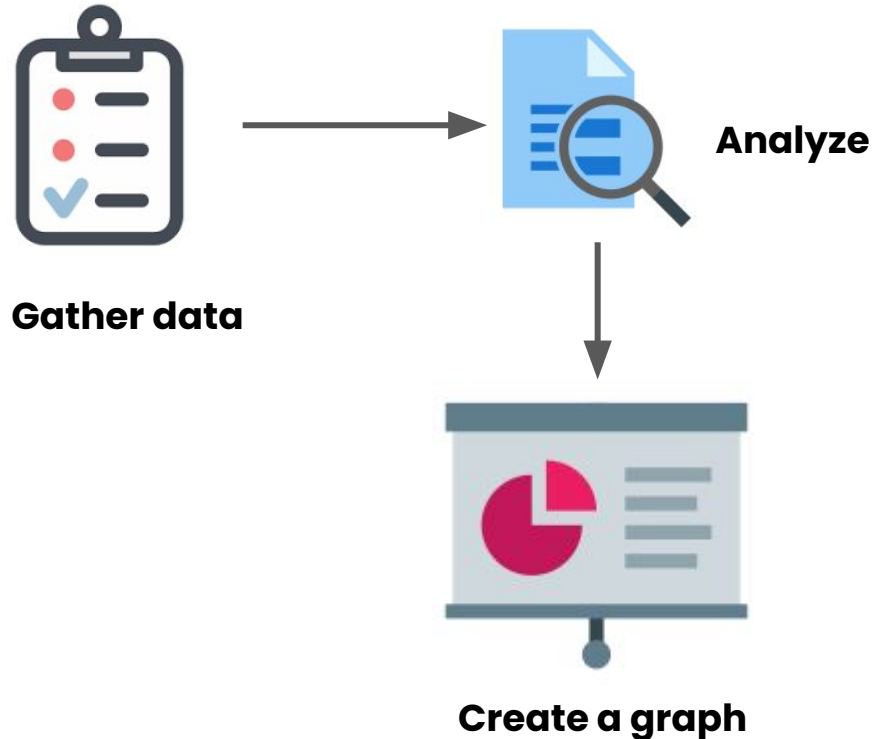
1. The basic idea of a function in programming.
2. How functions are used in Python.
3. To use operators to perform arithmetic operations.
4. How to call built-in Python functions.

Flow of data

Building blocks

Let's say you create a survey and then want to use the information to create a graph and show other people the results.

You might come up with a process like the one on the right.



Flow of data

In this example we're solving a problem by moving and transforming data. We get our data as input, then do some computations with it and then generate output.

This process is similar to the way we write functions in software.



Input



Processing



Output

Functions

Mathematical functions

Functions are something you may have seen in math class. They work in a similar way to those in programming.



Define: $f(x) = 2x$

$$f(2) = 4$$

$$f(6) = 12$$

In math we define a function once, but then use it multiple times for different values.

Functions

Python functions

Unlike functions in math, Python functions have more descriptive names and don't deal with just numbers.

Python functions look like math functions, except the names are more readable. When we run a function we say we are “calling” it.

```
print('Hi')  
max(1, 4, 3)  
int(5.6)  
float(6)  
min(1, 2, 3, 4, 5, 0)
```

Functions

Practice

Order of operations applies in programming mostly like in math, so remember to work your way from the inside out. Easy of these use the print function, so the last thing to run will output to the screen.

Remember:

input -> **processing** -> **output**

What is the output from:

1. `print(max(5, 4, 3))`
2. `print("")`
3. `print(int(6.555))`
4. `print(int(5.5))`
5. `print(float(min(4, 5, 3)))`
6. `print(min(5, max(1, 3)))`
7. `print(print('hi'))`

Pay close attention to `int(5.5)`

Arithmetic Operators

In Python, arithmetic operators are similar to those you've used in your math class, with a few additional options.

Those you'll be familiar with

- + Addition
- Subtraction
- / Division
- * Multiplication

Some not so familiar operators

- % Modulus
- // Floor division
- ** Exponent

Arithmetic Operators

The basics

Using basic operators and what you've learned about functions thus far, try and predict what the print statements to the right will output to the screen.

1. `print(5 * min(5, 6))`
2. `print(1 + 2 * 3 / 1 - 1)`
3. `print(max(5, 6 * 2) + 3)`
4. `print(int(3.1 * 2.01)/2.0)`
5. `print(max(1, 2) * max(5, 3))`
6. `print(int(5.5) * int(5.9))`

Arithmetic Operators

Something new

There are a few operators which you may not have seen before, but which you'll find quite useful. Take a look at the examples to the right and try to figure out how each of these work.

Modulus

$10 \% 2$ becomes 0

$13 \% 2$ becomes 1

$7 \% 5$ becomes 2

$9 \% 5$ becomes 4

Floor division

$10 // 2$ becomes 5

$13 // 2$ becomes 6

$7 // 5$ becomes 1

$14 // 5$ becomes 2

Exponent

$2 ** 3$ becomes 8

$3 ** 2$ becomes 9

$5 ** 3$ becomes 125

$4 ** 4$ becomes 256

A few more functions

More functions

There are many functions built into Python, some of which you'll use quite often. Below are a few more of them, with some examples to the right.

round, **abs**, and **pow**

1. `print(round(5.6))`
2. `print(round(5.41236, 4))`
3. `print(abs(-10))`
4. `print(pow(2, 3))`
5. `print(5 % 2)`
6. `print(16 // 3)`
7. `print(pow(min(1, 2), 4))`

Recap

We learned

1. How to **call** a function in Python.
2. How to use basic Python operators: **+ - * / % // ****
3. How to **print** information to the screen.
4. To use the **abs, pow, round, min, max, int** and **float** functions.